Top_Performers - best and second best, every data type and speed tier

Michael Hugh Harrington - 6 November 2026 - HX is Patent Pending (U.S. Provisional App. 64/192,843)

Bold = Michael Harrington's own software (HX, Harrington's Sets / hs_zstd, the cascade, HCM, the -hi tools and other modified or research builds) finishing first or second. Each cell shows the tool, its compression ratio (original size / compressed size) and, just under it, its decode and compress speed.

Speed range: tiers follow the 1-2-5 series of the Compression Tier Standard v2.0. M = LZ4's decode speed on that data type divided by the tool's decode speed. A tool counts in every tier whose limit it meets, so each tier means "this speed or faster"; the slowest tier is the overall best. All figures come from one machine and one session (same-machine campaign, 3-4 Nov 2026). HX combinations that depend on precomp, mzip or x264 are left out because they cannot ship. cmix and paq8px are not included.
Summary: across 290 type-and-tier categories (29 data types × 10 tiers), Michael Harrington's software is best in 120 and second best in 119.

1. Biased test files, 8 MB each

LZ4 decode speed on this type (M = 1): 354 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 177 MB/s	zstd -19 1.8657× decode 364.7 MB/s · comp 1.2 MB/s	FSE (tANS) 128 KB 1.7690× decode 433.0 MB/s · comp 239.7 MB/s
Tier 2: M ≤ 5 decode ≥ 71 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 3: M ≤ 10 decode ≥ 35 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 4: M ≤ 20 decode ≥ 18 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 5: M ≤ 50 decode ≥ 7.1 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 6: M ≤ 100 decode ≥ 3.5 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 7: M ≤ 200 decode ≥ 1.8 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.71 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**	**HX Complete** **2.0189×** **decode 148.1 MB/s · comp 0.4 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.18 MB/s	lpaq1 -9 2.0191× decode 0.4 MB/s · comp 0.4 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**
Tier 10: any speed (no limit)	lpaq1 -9 2.0191× decode 0.4 MB/s · comp 0.4 MB/s	**HX Fast** **2.0189×** **decode 152.5 MB/s · comp 0.6 MB/s**

2. NIST SP 800-90B real noise samples

LZ4 decode speed on this type (M = 1): 138 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 69 MB/s	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**	**HX lines + Sets** **2.4898×** **decode 304.3 MB/s · comp 4.1 MB/s**
Tier 2: M ≤ 5 decode ≥ 28 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 3: M ≤ 10 decode ≥ 14 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 4: M ≤ 20 decode ≥ 6.9 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 5: M ≤ 50 decode ≥ 2.8 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 6: M ≤ 100 decode ≥ 1.4 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.69 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.28 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.07 MB/s	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**
Tier 10: any speed (no limit)	**HX Complete** **2.4907×** **decode 54.3 MB/s · comp 0.3 MB/s**	**HX lines** **2.4898×** **decode 241.4 MB/s · comp 6.3 MB/s**

3. Random Turing machine output (uiq2, Generic Compression Benchmark data), 3 seeds

LZ4 decode speed on this type (M = 1): 291 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 145 MB/s	**HX Fast** **1.9207×** **decode 238.4 MB/s · comp 0.9 MB/s**	LZ4 (default) 1.2115× decode 291.8 MB/s · comp 195.5 MB/s
Tier 2: M ≤ 5 decode ≥ 58 MB/s	Brotli 11 1.9402× decode 91.4 MB/s · comp 0.4 MB/s	**HX Fast** **1.9207×** **decode 238.4 MB/s · comp 0.9 MB/s**
Tier 3: M ≤ 10 decode ≥ 29 MB/s	Brotli 11 1.9402× decode 91.4 MB/s · comp 0.4 MB/s	**HX Fast** **1.9207×** **decode 238.4 MB/s · comp 0.9 MB/s**
Tier 4: M ≤ 20 decode ≥ 15 MB/s	Brotli 11 1.9402× decode 91.4 MB/s · comp 0.4 MB/s	**HX Fast** **1.9207×** **decode 238.4 MB/s · comp 0.9 MB/s**
Tier 5: M ≤ 50 decode ≥ 5.8 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**	Brotli 11 1.9402× decode 91.4 MB/s · comp 0.4 MB/s
Tier 6: M ≤ 100 decode ≥ 2.9 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**	Brotli 11 1.9402× decode 91.4 MB/s · comp 0.4 MB/s
Tier 7: M ≤ 200 decode ≥ 1.5 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**	7-Zip PPMd (-mx9 -m0=PPMd) 2.0832× decode 2.3 MB/s · comp 2.3 MB/s
Tier 8: M ≤ 500 decode ≥ 0.58 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**	7-Zip PPMd (-mx9 -m0=PPMd) 2.0832× decode 2.3 MB/s · comp 2.3 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.15 MB/s	lpaq1 8 2.1341× decode 0.6 MB/s · comp 0.7 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**
Tier 10: any speed (no limit)	lpaq1 8 2.1341× decode 0.6 MB/s · comp 0.7 MB/s	**HX Complete** **2.1286×** **decode 7.2 MB/s · comp 0.7 MB/s**

4. Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR)

LZ4 decode speed on this type (M = 1): 61 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 30 MB/s	Brotli 11 7.9358× decode 61.2 MB/s · comp 0.4 MB/s	xz -9e 7.8701× decode 44.5 MB/s · comp 1.7 MB/s
Tier 2: M ≤ 5 decode ≥ 12 MB/s	**HX Complete + audio/colour lines** **8.6164×** **decode 19.3 MB/s · comp 0.2 MB/s**	BSC 8.5806× decode 19.0 MB/s · comp 16.3 MB/s
Tier 3: M ≤ 10 decode ≥ 6.1 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s	**HX Complete + audio/colour lines** **8.6164×** **decode 19.3 MB/s · comp 0.2 MB/s**
Tier 4: M ≤ 20 decode ≥ 3.0 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s	**HX Complete + audio/colour lines** **8.6164×** **decode 19.3 MB/s · comp 0.2 MB/s**
Tier 5: M ≤ 50 decode ≥ 1.2 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s	**HX Complete + audio/colour lines** **8.6164×** **decode 19.3 MB/s · comp 0.2 MB/s**
Tier 6: M ≤ 100 decode ≥ 0.61 MB/s	kanzi -l 9 10.0133× decode 0.7 MB/s · comp 0.6 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s
Tier 7: M ≤ 200 decode ≥ 0.30 MB/s	kanzi -l 9 10.0133× decode 0.7 MB/s · comp 0.6 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s
Tier 8: M ≤ 500 decode ≥ 0.12 MB/s	kanzi -l 9 10.0133× decode 0.7 MB/s · comp 0.6 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.03 MB/s	kanzi -l 9 10.0133× decode 0.7 MB/s · comp 0.6 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s
Tier 10: any speed (no limit)	kanzi -l 9 10.0133× decode 0.7 MB/s · comp 0.6 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 8.6863× decode 7.7 MB/s · comp 8.0 MB/s

5. Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc++, cc1plus, git)

LZ4 decode speed on this type (M = 1): 347 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 173 MB/s	**HX Fast + audio/colour lines** **3.0427×** **decode 201.6 MB/s · comp 0.4 MB/s**	zstd -19 2.8333× decode 225.4 MB/s · comp 2.1 MB/s
Tier 2: M ≤ 5 decode ≥ 69 MB/s	**HX Fast + audio/colour lines** **3.0427×** **decode 201.6 MB/s · comp 0.4 MB/s**	Brotli 11 2.9804× decode 123.3 MB/s · comp 0.3 MB/s
Tier 3: M ≤ 10 decode ≥ 35 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s	**HX Fast + audio/colour lines** **3.0427×** **decode 201.6 MB/s · comp 0.4 MB/s**
Tier 4: M ≤ 20 decode ≥ 17 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s	**HX Fast + audio/colour lines** **3.0427×** **decode 201.6 MB/s · comp 0.4 MB/s**
Tier 5: M ≤ 50 decode ≥ 6.9 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s	**HX Fast + audio/colour lines** **3.0427×** **decode 201.6 MB/s · comp 0.4 MB/s**
Tier 6: M ≤ 100 decode ≥ 3.5 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 3.2774× decode 4.5 MB/s · comp 4.4 MB/s
Tier 7: M ≤ 200 decode ≥ 1.7 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 3.2774× decode 4.5 MB/s · comp 4.4 MB/s
Tier 8: M ≤ 500 decode ≥ 0.69 MB/s	kanzi -l 9 3.9559× decode 1.3 MB/s · comp 1.1 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.17 MB/s	kanzi -l 9 3.9559× decode 1.3 MB/s · comp 1.1 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s
Tier 10: any speed (no limit)	kanzi -l 9 3.9559× decode 1.3 MB/s · comp 1.1 MB/s	7-Zip -mx9 (LZMA2) 3.4958× decode 41.3 MB/s · comp 4.2 MB/s

6. Silesia, 12 files

LZ4 decode speed on this type (M = 1): 482 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 241 MB/s	**HX Fast** **4.2579×** **decode 308.9 MB/s · comp 0.9 MB/s**	LZ4 (default) 2.1121× decode 482.8 MB/s · comp 311.7 MB/s
Tier 2: M ≤ 5 decode ≥ 96 MB/s	Brotli 11 4.2760× decode 176.2 MB/s · comp 0.4 MB/s	**HX Fast** **4.2579×** **decode 308.9 MB/s · comp 0.9 MB/s**
Tier 3: M ≤ 10 decode ≥ 48 MB/s	7-Zip -mx9 (LZMA2) 4.3467× decode 62.2 MB/s · comp 2.9 MB/s	Brotli 11 4.2760× decode 176.2 MB/s · comp 0.4 MB/s
Tier 4: M ≤ 20 decode ≥ 24 MB/s	**HX Complete** **4.7243×** **decode 24.6 MB/s · comp 0.8 MB/s**	7-Zip -mx9 (LZMA2) 4.3467× decode 62.2 MB/s · comp 2.9 MB/s
Tier 5: M ≤ 50 decode ≥ 9.6 MB/s	**HX Complete** **4.7243×** **decode 24.6 MB/s · comp 0.8 MB/s**	BSC 4.5542× decode 15.5 MB/s · comp 11.7 MB/s
Tier 6: M ≤ 100 decode ≥ 4.8 MB/s	mzip 4.9178× decode 8.6 MB/s · comp 0.1 MB/s	**HX Complete** **4.7243×** **decode 24.6 MB/s · comp 0.8 MB/s**
Tier 7: M ≤ 200 decode ≥ 2.4 MB/s	mzip 4.9178× decode 8.6 MB/s · comp 0.1 MB/s	**HX Complete** **4.7243×** **decode 24.6 MB/s · comp 0.8 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.96 MB/s	kanzi -l 9 5.1964× decode 2.2 MB/s · comp 2.0 MB/s	mzip 4.9178× decode 8.6 MB/s · comp 0.1 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.24 MB/s	kanzi -l 9 5.1964× decode 2.2 MB/s · comp 2.0 MB/s	mzip 4.9178× decode 8.6 MB/s · comp 0.1 MB/s
Tier 10: any speed (no limit)	kanzi -l 9 5.1964× decode 2.2 MB/s · comp 2.0 MB/s	mzip 4.9178× decode 8.6 MB/s · comp 0.1 MB/s

7. Calgary corpus, 18 files

LZ4 decode speed on this type (M = 1): 39 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 19 MB/s	Brotli 11 3.7949× decode 41.2 MB/s · comp 0.5 MB/s	**HX Fast** **3.5582×** **decode 40.6 MB/s · comp 0.4 MB/s**
Tier 2: M ≤ 5 decode ≥ 7.7 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**	BSC 4.0359× decode 9.6 MB/s · comp 8.3 MB/s
Tier 3: M ≤ 10 decode ≥ 3.9 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**	bzip3 4.0491× decode 7.4 MB/s · comp 6.8 MB/s
Tier 4: M ≤ 20 decode ≥ 1.9 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**	bzip3 4.0491× decode 7.4 MB/s · comp 6.8 MB/s
Tier 5: M ≤ 50 decode ≥ 0.77 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**	bzip3 4.0491× decode 7.4 MB/s · comp 6.8 MB/s
Tier 6: M ≤ 100 decode ≥ 0.39 MB/s	kanzi -l 9 4.4439× decode 0.4 MB/s · comp 0.3 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.19 MB/s	kanzi -l 9 4.4439× decode 0.4 MB/s · comp 0.3 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.08 MB/s	kanzi -l 9 4.4439× decode 0.4 MB/s · comp 0.3 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.02 MB/s	kanzi -l 9 4.4439× decode 0.4 MB/s · comp 0.3 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**
Tier 10: any speed (no limit)	kanzi -l 9 4.4439× decode 0.4 MB/s · comp 0.3 MB/s	**HX Complete** **4.0808×** **decode 10.3 MB/s · comp 0.4 MB/s**

8. Canterbury corpus, 11 files

LZ4 decode speed on this type (M = 1): 51 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 26 MB/s	Brotli 11 5.7275× decode 55.1 MB/s · comp 0.5 MB/s	**HX Fast** **5.4423×** **decode 57.4 MB/s · comp 0.4 MB/s**
Tier 2: M ≤ 5 decode ≥ 10 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**	BSC 6.1437× decode 12.8 MB/s · comp 11.7 MB/s
Tier 3: M ≤ 10 decode ≥ 5.1 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**	BSC 6.1437× decode 12.8 MB/s · comp 11.7 MB/s
Tier 4: M ≤ 20 decode ≥ 2.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**	BSC 6.1437× decode 12.8 MB/s · comp 11.7 MB/s
Tier 5: M ≤ 50 decode ≥ 1.0 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**	BSC 6.1437× decode 12.8 MB/s · comp 11.7 MB/s
Tier 6: M ≤ 100 decode ≥ 0.51 MB/s	kanzi -l 9 6.8138× decode 0.6 MB/s · comp 0.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.26 MB/s	kanzi -l 9 6.8138× decode 0.6 MB/s · comp 0.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.10 MB/s	kanzi -l 9 6.8138× decode 0.6 MB/s · comp 0.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.03 MB/s	kanzi -l 9 6.8138× decode 0.6 MB/s · comp 0.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**
Tier 10: any speed (no limit)	kanzi -l 9 6.8138× decode 0.6 MB/s · comp 0.6 MB/s	**HX Complete** **6.3256×** **decode 16.1 MB/s · comp 0.4 MB/s**

9. Canterbury large corpus, 3 files

LZ4 decode speed on this type (M = 1): 233 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 117 MB/s	Brotli 11 4.4600× decode 146.8 MB/s · comp 0.3 MB/s	**HX Fast** **4.3809×** **decode 328.2 MB/s · comp 0.6 MB/s**
Tier 2: M ≤ 5 decode ≥ 47 MB/s	Brotli 11 4.4600× decode 146.8 MB/s · comp 0.3 MB/s	**HX Fast** **4.3809×** **decode 328.2 MB/s · comp 0.6 MB/s**
Tier 3: M ≤ 10 decode ≥ 23 MB/s	Brotli 11 4.4600× decode 146.8 MB/s · comp 0.3 MB/s	**HX Fast** **4.3809×** **decode 328.2 MB/s · comp 0.6 MB/s**
Tier 4: M ≤ 20 decode ≥ 12 MB/s	BSC 4.9406× decode 17.0 MB/s · comp 13.0 MB/s	Brotli 11 4.4600× decode 146.8 MB/s · comp 0.3 MB/s
Tier 5: M ≤ 50 decode ≥ 4.7 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**	bzip3 4.9480× decode 6.9 MB/s · comp 7.6 MB/s
Tier 6: M ≤ 100 decode ≥ 2.3 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**	bzip3 4.9480× decode 6.9 MB/s · comp 7.6 MB/s
Tier 7: M ≤ 200 decode ≥ 1.2 MB/s	kanzi -l 9 5.1603× decode 2.2 MB/s · comp 1.8 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.47 MB/s	kanzi -l 9 5.1603× decode 2.2 MB/s · comp 1.8 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.12 MB/s	kanzi -l 9 5.1603× decode 2.2 MB/s · comp 1.8 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**
Tier 10: any speed (no limit)	kanzi -l 9 5.1603× decode 2.2 MB/s · comp 1.8 MB/s	**HX Complete** **4.9604×** **decode 8.0 MB/s · comp 0.5 MB/s**

10. Synthetic test files, 5 files

LZ4 decode speed on this type (M = 1): 538 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 269 MB/s	**HX Fast** **1.6410×** **decode 330.2 MB/s · comp 1.3 MB/s**	LZ4 (default) 1.1128× decode 539.0 MB/s · comp 331.6 MB/s
Tier 2: M ≤ 5 decode ≥ 108 MB/s	**HX Fast** **1.6410×** **decode 330.2 MB/s · comp 1.3 MB/s**	Brotli 11 1.5756× decode 128.6 MB/s · comp 0.3 MB/s
Tier 3: M ≤ 10 decode ≥ 54 MB/s	**HX Fast** **1.6410×** **decode 330.2 MB/s · comp 1.3 MB/s**	Brotli 11 1.5756× decode 128.6 MB/s · comp 0.3 MB/s
Tier 4: M ≤ 20 decode ≥ 27 MB/s	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s	**HX Fast** **1.6410×** **decode 330.2 MB/s · comp 1.3 MB/s**
Tier 5: M ≤ 50 decode ≥ 11 MB/s	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s
Tier 6: M ≤ 100 decode ≥ 5.4 MB/s	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s
Tier 7: M ≤ 200 decode ≥ 2.7 MB/s	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s
Tier 8: M ≤ 500 decode ≥ 1.1 MB/s	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.27 MB/s	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s
Tier 10: any speed (no limit)	**HX Complete** **1.6642×** **decode 22.2 MB/s · comp 0.7 MB/s**	7-Zip -mx9 (LZMA2) 1.6479× decode 44.7 MB/s · comp 6.9 MB/s

11. Speech, WAVs, 8 kHz mono (FSDD)

LZ4 decode speed on this type (M = 1): 308 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 154 MB/s	zstd -19 1.4319× decode 196.2 MB/s · comp 2.7 MB/s	LZ4 (default) 1.0963× decode 308.8 MB/s · comp 233.3 MB/s
Tier 2: M ≤ 5 decode ≥ 62 MB/s	WavPack -hhx6 (official, max) 1.9753× decode 63.2 MB/s · comp 1.0 MB/s	**HX Fast + audio/colour lines** **1.9117×** **decode 101.9 MB/s · comp 0.3 MB/s**
Tier 3: M ≤ 10 decode ≥ 31 MB/s	WavPack -hhx6 (official, max) 1.9753× decode 63.2 MB/s · comp 1.0 MB/s	WavPack -hh (official) 1.9371× decode 50.8 MB/s · comp 28.8 MB/s
Tier 4: M ≤ 20 decode ≥ 15 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s	WavPack (ffmpeg, level 8) 1.9787× decode 23.3 MB/s · comp 0.1 MB/s
Tier 5: M ≤ 50 decode ≥ 6.2 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s	WavPack (ffmpeg, level 8) 1.9787× decode 23.3 MB/s · comp 0.1 MB/s
Tier 6: M ≤ 100 decode ≥ 3.1 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s	WavPack (ffmpeg, level 8) 1.9787× decode 23.3 MB/s · comp 0.1 MB/s
Tier 7: M ≤ 200 decode ≥ 1.5 MB/s	OptimFROG --preset max 2.0743× decode 2.6 MB/s · comp 0.5 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s
Tier 8: M ≤ 500 decode ≥ 0.62 MB/s	OptimFROG --preset max 2.0743× decode 2.6 MB/s · comp 0.5 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.15 MB/s	OptimFROG --preset max 2.0743× decode 2.6 MB/s · comp 0.5 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s
Tier 10: any speed (no limit)	OptimFROG --preset max 2.0743× decode 2.6 MB/s · comp 0.5 MB/s	OptimFROG --preset 2 2.0420× decode 19.9 MB/s · comp 12.5 MB/s

12. Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data)

LZ4 decode speed on this type (M = 1): 447 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 224 MB/s	zstd -19 1.2119× decode 264.8 MB/s · comp 2.9 MB/s	LZ4 (default) 1.0513× decode 447.0 MB/s · comp 415.7 MB/s
Tier 2: M ≤ 5 decode ≥ 89 MB/s	flac -8 -e -p (max) 2.6090× decode 128.0 MB/s · comp 1.7 MB/s	flac -8 2.5998× decode 126.5 MB/s · comp 42.7 MB/s
Tier 3: M ≤ 10 decode ≥ 45 MB/s	WavPack -hhx6 (official, max) 2.6174× decode 63.2 MB/s · comp 1.0 MB/s	flac -8 -e -p (max) 2.6090× decode 128.0 MB/s · comp 1.7 MB/s
Tier 4: M ≤ 20 decode ≥ 22 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s	WavPack -hhx6 (official, max) 2.6174× decode 63.2 MB/s · comp 1.0 MB/s
Tier 5: M ≤ 50 decode ≥ 8.9 MB/s	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s
Tier 6: M ≤ 100 decode ≥ 4.5 MB/s	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s
Tier 7: M ≤ 200 decode ≥ 2.2 MB/s	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s
Tier 8: M ≤ 500 decode ≥ 0.89 MB/s	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.22 MB/s	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s
Tier 10: any speed (no limit)	Monkey's Audio -c5000 2.8524× decode 18.9 MB/s · comp 18.3 MB/s	Monkey's Audio -c4000 2.8483× decode 42.3 MB/s · comp 7.7 MB/s

13. Environmental recordings, WAVs, 44.1 kHz mono (ESC-50)

LZ4 decode speed on this type (M = 1): 102 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 51 MB/s	flac -8 -e -p (max) 2.2176× decode 78.2 MB/s · comp 3.0 MB/s	flac -8 2.2126× decode 79.2 MB/s · comp 48.5 MB/s
Tier 2: M ≤ 5 decode ≥ 20 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 3: M ≤ 10 decode ≥ 10 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 4: M ≤ 20 decode ≥ 5.1 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 5: M ≤ 50 decode ≥ 2.0 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 6: M ≤ 100 decode ≥ 1.0 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.51 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.20 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.05 MB/s	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**
Tier 10: any speed (no limit)	**HX Fast + audio/colour lines** **2.3531×** **decode 38.5 MB/s · comp 0.2 MB/s**	**HX Complete + audio/colour lines** **2.3531×** **decode 38.0 MB/s · comp 0.2 MB/s**

14. Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC

LZ4 decode speed on this type (M = 1): 194 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 97 MB/s	Brotli 11 1.0108× decode 105.1 MB/s · comp 0.6 MB/s	zstd -19 1.0093× decode 209.3 MB/s · comp 5.8 MB/s
Tier 2: M ≤ 5 decode ≥ 39 MB/s	Brotli 11 1.0108× decode 105.1 MB/s · comp 0.6 MB/s	zstd -19 1.0093× decode 209.3 MB/s · comp 5.8 MB/s
Tier 3: M ≤ 10 decode ≥ 19 MB/s	Brotli 11 1.0108× decode 105.1 MB/s · comp 0.6 MB/s	zstd -19 1.0093× decode 209.3 MB/s · comp 5.8 MB/s
Tier 4: M ≤ 20 decode ≥ 9.7 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	Brotli 11 1.0108× decode 105.1 MB/s · comp 0.6 MB/s
Tier 5: M ≤ 50 decode ≥ 3.9 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**
Tier 6: M ≤ 100 decode ≥ 1.9 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.97 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.39 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.10 MB/s	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**
Tier 10: any speed (no limit)	precomp (lzma) 1.0251× decode 13.5 MB/s · comp 2.5 MB/s	**HX Complete** **1.0129×** **decode 7.9 MB/s · comp 0.7 MB/s**

15. Colour photos, raw RGB (Kodak 24, PPM)

LZ4 decode speed on this type (M = 1): 193 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 96 MB/s	zstd -19 1.8638× decode 153.0 MB/s · comp 3.4 MB/s	LZ4 (default) 1.1796× decode 192.6 MB/s · comp 157.3 MB/s
Tier 2: M ≤ 5 decode ≥ 39 MB/s	WebP lossless (z9) 2.5269× decode 78.6 MB/s · comp 0.1 MB/s	**HX Fast** **1.9535×** **decode 83.3 MB/s · comp 0.5 MB/s**
Tier 3: M ≤ 10 decode ≥ 19 MB/s	WebP lossless (z9) 2.5269× decode 78.6 MB/s · comp 0.1 MB/s	7-Zip -mx9 (LZMA2) 1.9574× decode 26.1 MB/s · comp 6.8 MB/s
Tier 4: M ≤ 20 decode ≥ 9.6 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 5: M ≤ 50 decode ≥ 3.9 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 6: M ≤ 100 decode ≥ 1.9 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 7: M ≤ 200 decode ≥ 0.96 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 8: M ≤ 500 decode ≥ 0.39 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.10 MB/s	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s
Tier 10: any speed (no limit)	**HX Complete + specialists** **2.8321×** **decode 13.6 MB/s · comp 0.1 MB/s**	JPEG XL lossless (e9) 2.8288× decode 13.8 MB/s · comp 0.5 MB/s

16. Rendered document pages / screenshots, raw (PGM/PPM)

LZ4 decode speed on this type (M = 1): 308 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 154 MB/s	xz -9e 33.1323× decode 171.0 MB/s · comp 3.0 MB/s	Brotli 11 32.7782× decode 242.4 MB/s · comp 1.3 MB/s
Tier 2: M ≤ 5 decode ≥ 62 MB/s	**HX Complete + specialists** **34.8334×** **decode 68.5 MB/s · comp 0.3 MB/s**	BSC 34.4585× decode 75.0 MB/s · comp 68.9 MB/s
Tier 3: M ≤ 10 decode ≥ 31 MB/s	**HX Complete + specialists** **34.8334×** **decode 68.5 MB/s · comp 0.3 MB/s**	**HX Complete** **34.7865×** **decode 60.4 MB/s · comp 0.9 MB/s**
Tier 4: M ≤ 20 decode ≥ 15 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	**HX Complete + specialists** **34.8334×** **decode 68.5 MB/s · comp 0.3 MB/s**
Tier 5: M ≤ 50 decode ≥ 6.2 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	**HX Complete + specialists** **34.8334×** **decode 68.5 MB/s · comp 0.3 MB/s**
Tier 6: M ≤ 100 decode ≥ 3.1 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	kanzi -l 9 35.3289× decode 4.0 MB/s · comp 3.5 MB/s
Tier 7: M ≤ 200 decode ≥ 1.5 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	kanzi -l 9 35.3289× decode 4.0 MB/s · comp 3.5 MB/s
Tier 8: M ≤ 500 decode ≥ 0.62 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	kanzi -l 9 35.3289× decode 4.0 MB/s · comp 3.5 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.15 MB/s	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	kanzi -l 9 35.3289× decode 4.0 MB/s · comp 3.5 MB/s
Tier 10: any speed (no limit)	7-Zip PPMd (-mx9 -m0=PPMd) 36.2243× decode 29.8 MB/s · comp 33.9 MB/s	kanzi -l 9 35.3289× decode 4.0 MB/s · comp 3.5 MB/s

17. Medical DICOM files (imageio test images)

LZ4 decode speed on this type (M = 1): 99 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 50 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 2: M ≤ 5 decode ≥ 20 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 3: M ≤ 10 decode ≥ 9.9 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 4: M ≤ 20 decode ≥ 5.0 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 5: M ≤ 50 decode ≥ 2.0 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 6: M ≤ 100 decode ≥ 0.99 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.50 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.20 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.05 MB/s	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**
Tier 10: any speed (no limit)	**HX Fast** **3.1517×** **decode 70.9 MB/s · comp 0.6 MB/s**	**HX Complete** **3.1517×** **decode 70.9 MB/s · comp 0.5 MB/s**

18. PNG files (Kodak 24)

LZ4 decode speed on this type (M = 1): 132 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 66 MB/s	**HX Fast** **1.0004×** **decode 149.5 MB/s · comp 0.7 MB/s**	gzip -9 1.0003× decode 81.5 MB/s · comp 14.0 MB/s
Tier 2: M ≤ 5 decode ≥ 26 MB/s	7-Zip -mx9 (LZMA2) 1.0016× decode 33.2 MB/s · comp 11.2 MB/s	xz -9e 1.0015× decode 40.9 MB/s · comp 2.6 MB/s
Tier 3: M ≤ 10 decode ≥ 13 MB/s	7-Zip -mx9 (LZMA2) 1.0016× decode 33.2 MB/s · comp 11.2 MB/s	xz -9e 1.0015× decode 40.9 MB/s · comp 2.6 MB/s
Tier 4: M ≤ 20 decode ≥ 6.6 MB/s	**HX Complete** **1.0087×** **decode 9.3 MB/s · comp 0.5 MB/s**	BSC 1.0062× decode 6.8 MB/s · comp 5.6 MB/s
Tier 5: M ≤ 50 decode ≥ 2.6 MB/s	**HX Complete** **1.0087×** **decode 9.3 MB/s · comp 0.5 MB/s**	bzip3 1.0078× decode 6.0 MB/s · comp 5.6 MB/s
Tier 6: M ≤ 100 decode ≥ 1.3 MB/s	**H-Deflate + HX Complete + image routes** **1.5390×** **decode 1.6 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.0808× decode 1.9 MB/s · comp 0.6 MB/s
Tier 7: M ≤ 200 decode ≥ 0.66 MB/s	**H-Deflate + HX Complete + image routes** **1.5390×** **decode 1.6 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.0808× decode 1.9 MB/s · comp 0.6 MB/s
Tier 8: M ≤ 500 decode ≥ 0.26 MB/s	**H-Deflate + HX Complete + image routes** **1.5390×** **decode 1.6 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.0808× decode 1.9 MB/s · comp 0.6 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.07 MB/s	**H-Deflate + HX Complete + image routes** **1.5390×** **decode 1.6 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.0808× decode 1.9 MB/s · comp 0.6 MB/s
Tier 10: any speed (no limit)	**H-Deflate + HX Complete + image routes** **1.5390×** **decode 1.6 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.0808× decode 1.9 MB/s · comp 0.6 MB/s

19. JPEG photos

LZ4 decode speed on this type (M = 1): 251 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 125 MB/s	**HX Fast** **1.0095×** **decode 152.9 MB/s · comp 0.9 MB/s**	zstd -19 1.0076× decode 231.0 MB/s · comp 5.5 MB/s
Tier 2: M ≤ 5 decode ≥ 50 MB/s	**HX Fast** **1.0095×** **decode 152.9 MB/s · comp 0.9 MB/s**	zstd -19 1.0076× decode 231.0 MB/s · comp 5.5 MB/s
Tier 3: M ≤ 10 decode ≥ 25 MB/s	**HX Fast** **1.0095×** **decode 152.9 MB/s · comp 0.9 MB/s**	zstd -19 1.0076× decode 231.0 MB/s · comp 5.5 MB/s
Tier 4: M ≤ 20 decode ≥ 13 MB/s	**HX Fast** **1.0095×** **decode 152.9 MB/s · comp 0.9 MB/s**	zstd -19 1.0076× decode 231.0 MB/s · comp 5.5 MB/s
Tier 5: M ≤ 50 decode ≥ 5.0 MB/s	Lepton 1.2760× decode 7.5 MB/s · comp 2.3 MB/s	precomp (lzma) 1.2716× decode 7.2 MB/s · comp 2.4 MB/s
Tier 6: M ≤ 100 decode ≥ 2.5 MB/s	Lepton 1.2760× decode 7.5 MB/s · comp 2.3 MB/s	precomp (lzma) 1.2716× decode 7.2 MB/s · comp 2.4 MB/s
Tier 7: M ≤ 200 decode ≥ 1.3 MB/s	packJPG 1.2848× decode 2.3 MB/s · comp 2.1 MB/s	**HX Complete + specialists** **1.2847×** **decode 2.3 MB/s · comp 0.2 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.50 MB/s	packJPG 1.2848× decode 2.3 MB/s · comp 2.1 MB/s	**HX Complete + specialists** **1.2847×** **decode 2.3 MB/s · comp 0.2 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.13 MB/s	packJPG 1.2848× decode 2.3 MB/s · comp 2.1 MB/s	**HX Complete + specialists** **1.2847×** **decode 2.3 MB/s · comp 0.2 MB/s**
Tier 10: any speed (no limit)	packJPG 1.2848× decode 2.3 MB/s · comp 2.1 MB/s	**HX Complete + specialists** **1.2847×** **decode 2.3 MB/s · comp 0.2 MB/s**

20. GIF and TIFF files

LZ4 decode speed on this type (M = 1): 113 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 57 MB/s	Brotli 11 1.2878× decode 60.1 MB/s · comp 0.5 MB/s	**HX Fast** **1.2358×** **decode 75.5 MB/s · comp 0.9 MB/s**
Tier 2: M ≤ 5 decode ≥ 23 MB/s	Brotli 11 1.2878× decode 60.1 MB/s · comp 0.5 MB/s	**HX Fast** **1.2358×** **decode 75.5 MB/s · comp 0.9 MB/s**
Tier 3: M ≤ 10 decode ≥ 11 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	Brotli 11 1.2878× decode 60.1 MB/s · comp 0.5 MB/s
Tier 4: M ≤ 20 decode ≥ 5.7 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 5: M ≤ 50 decode ≥ 2.3 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 6: M ≤ 100 decode ≥ 1.1 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 7: M ≤ 200 decode ≥ 0.57 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 8: M ≤ 500 decode ≥ 0.23 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.06 MB/s	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**
Tier 10: any speed (no limit)	precomp (lzma) 1.3876× decode 14.4 MB/s · comp 2.0 MB/s	**HX Complete** **1.2992×** **decode 6.1 MB/s · comp 0.6 MB/s**

21. Raw video (Y4M; frames decoded from H.264 sources + one raw test clip)

LZ4 decode speed on this type (M = 1): 692 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 346 MB/s	zstd -19 6.8305× decode 390.9 MB/s · comp 1.8 MB/s	LZ4 (default) 3.2183× decode 691.4 MB/s · comp 459.3 MB/s
Tier 2: M ≤ 5 decode ≥ 138 MB/s	Brotli 11 7.3681× decode 245.9 MB/s · comp 0.4 MB/s	zstd -19 6.8305× decode 390.9 MB/s · comp 1.8 MB/s
Tier 3: M ≤ 10 decode ≥ 69 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	FFV1 (level 3) 11.5804× decode 80.0 MB/s · comp 79.9 MB/s
Tier 4: M ≤ 20 decode ≥ 35 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 5: M ≤ 50 decode ≥ 14 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 6: M ≤ 100 decode ≥ 6.9 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 7: M ≤ 200 decode ≥ 3.5 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 8: M ≤ 500 decode ≥ 1.4 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.35 MB/s	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**
Tier 10: any speed (no limit)	x264 lossless (veryslow) 16.8571× decode 110.7 MB/s · comp 14.0 MB/s	**HX Complete + specialists** **11.5839×** **decode 66.4 MB/s · comp 0.7 MB/s**

22. Compressed video files (H.264, HEVC, VP9)

LZ4 decode speed on this type (M = 1): 72 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 36 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 2: M ≤ 5 decode ≥ 14 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 3: M ≤ 10 decode ≥ 7.2 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 4: M ≤ 20 decode ≥ 3.6 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 5: M ≤ 50 decode ≥ 1.4 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 6: M ≤ 100 decode ≥ 0.72 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	zstd -19 1.0417× decode 53.2 MB/s · comp 6.1 MB/s
Tier 7: M ≤ 200 decode ≥ 0.36 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	kanzi -l 9 1.0419× decode 0.6 MB/s · comp 0.4 MB/s
Tier 8: M ≤ 500 decode ≥ 0.14 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	kanzi -l 9 1.0419× decode 0.6 MB/s · comp 0.4 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.04 MB/s	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	kanzi -l 9 1.0419× decode 0.6 MB/s · comp 0.4 MB/s
Tier 10: any speed (no limit)	Brotli 11 1.0438× decode 55.6 MB/s · comp 0.4 MB/s	kanzi -l 9 1.0419× decode 0.6 MB/s · comp 0.4 MB/s

23. PDF files from the pdf.js test set

LZ4 decode speed on this type (M = 1): 129 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 65 MB/s	Brotli 11 1.3404× decode 74.8 MB/s · comp 0.4 MB/s	**HX Fast** **1.3304×** **decode 177.2 MB/s · comp 0.7 MB/s**
Tier 2: M ≤ 5 decode ≥ 26 MB/s	Brotli 11 1.3404× decode 74.8 MB/s · comp 0.4 MB/s	7-Zip -mx9 (LZMA2) 1.3370× decode 27.1 MB/s · comp 8.8 MB/s
Tier 3: M ≤ 10 decode ≥ 13 MB/s	**HX Complete** **1.3531×** **decode 20.4 MB/s · comp 0.5 MB/s**	Brotli 11 1.3404× decode 74.8 MB/s · comp 0.4 MB/s
Tier 4: M ≤ 20 decode ≥ 6.5 MB/s	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s	**HX Complete** **1.3531×** **decode 20.4 MB/s · comp 0.5 MB/s**
Tier 5: M ≤ 50 decode ≥ 2.6 MB/s	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s
Tier 6: M ≤ 100 decode ≥ 1.3 MB/s	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s
Tier 7: M ≤ 200 decode ≥ 0.65 MB/s	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s
Tier 8: M ≤ 500 decode ≥ 0.26 MB/s	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.06 MB/s	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s
Tier 10: any speed (no limit)	**H-Deflate + HX Complete + image routes** **1.6861×** **decode 6.3 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.6576× decode 8.3 MB/s · comp 1.1 MB/s

24. Packet captures, public pcapng files

LZ4 decode speed on this type (M = 1): 90 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 45 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	**HX Fast** **1.3664×** **decode 103.5 MB/s · comp 0.9 MB/s**
Tier 2: M ≤ 5 decode ≥ 18 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	**HX Fast** **1.3664×** **decode 103.5 MB/s · comp 0.9 MB/s**
Tier 3: M ≤ 10 decode ≥ 9.0 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	xz -9e 1.3854× decode 15.7 MB/s · comp 1.9 MB/s
Tier 4: M ≤ 20 decode ≥ 4.5 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	xz -9e 1.3854× decode 15.7 MB/s · comp 1.9 MB/s
Tier 5: M ≤ 50 decode ≥ 1.8 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	xz -9e 1.3854× decode 15.7 MB/s · comp 1.9 MB/s
Tier 6: M ≤ 100 decode ≥ 0.90 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s	xz -9e 1.3854× decode 15.7 MB/s · comp 1.9 MB/s
Tier 7: M ≤ 200 decode ≥ 0.45 MB/s	kanzi -l 9 1.4192× decode 0.5 MB/s · comp 0.5 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s
Tier 8: M ≤ 500 decode ≥ 0.18 MB/s	kanzi -l 9 1.4192× decode 0.5 MB/s · comp 0.5 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.05 MB/s	kanzi -l 9 1.4192× decode 0.5 MB/s · comp 0.5 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s
Tier 10: any speed (no limit)	kanzi -l 9 1.4192× decode 0.5 MB/s · comp 0.5 MB/s	Brotli 11 1.3976× decode 58.0 MB/s · comp 0.3 MB/s

25. Packet capture, one 113 MB merged pcap

LZ4 decode speed on this type (M = 1): 855 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 428 MB/s	**HX Complete** **1.3775×** **decode 464.7 MB/s · comp 0.7 MB/s**	LZ4 (default) 1.1484× decode 855.4 MB/s · comp 868.6 MB/s
Tier 2: M ≤ 5 decode ≥ 171 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**	**HX Complete** **1.3775×** **decode 464.7 MB/s · comp 0.7 MB/s**
Tier 3: M ≤ 10 decode ≥ 86 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**	**HX Complete** **1.3775×** **decode 464.7 MB/s · comp 0.7 MB/s**
Tier 4: M ≤ 20 decode ≥ 43 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**	**HX Complete** **1.3775×** **decode 464.7 MB/s · comp 0.7 MB/s**
Tier 5: M ≤ 50 decode ≥ 17 MB/s	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**
Tier 6: M ≤ 100 decode ≥ 8.6 MB/s	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**
Tier 7: M ≤ 200 decode ≥ 4.3 MB/s	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**
Tier 8: M ≤ 500 decode ≥ 1.7 MB/s	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**
Tier 9: M ≤ 2000 decode ≥ 0.43 MB/s	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**
Tier 10: any speed (no limit)	7-Zip -mx9 (LZMA2) 1.3862× decode 19.4 MB/s · comp 8.1 MB/s	**HX Fast** **1.3775×** **decode 418.2 MB/s · comp 1.2 MB/s**

26. Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data)

LZ4 decode speed on this type (M = 1): 35 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 18 MB/s	Brotli 11 1.5314× decode 32.3 MB/s · comp 0.4 MB/s	xz -9e 1.5276× decode 19.2 MB/s · comp 1.7 MB/s
Tier 2: M ≤ 5 decode ≥ 7.1 MB/s	**HX Complete** **1.5328×** **decode 14.3 MB/s · comp 0.5 MB/s**	Brotli 11 1.5314× decode 32.3 MB/s · comp 0.4 MB/s
Tier 3: M ≤ 10 decode ≥ 3.5 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 4: M ≤ 20 decode ≥ 1.8 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 5: M ≤ 50 decode ≥ 0.71 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 6: M ≤ 100 decode ≥ 0.35 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 7: M ≤ 200 decode ≥ 0.18 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 8: M ≤ 500 decode ≥ 0.07 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.02 MB/s	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s
Tier 10: any speed (no limit)	**H-Deflate + HX Complete + image routes** **2.0581×** **decode 4.5 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.9213× decode 6.4 MB/s · comp 0.8 MB/s

27. SQLite database (Chinook)

LZ4 decode speed on this type (M = 1): 167 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 84 MB/s	Brotli 11 4.3532× decode 84.0 MB/s · comp 0.2 MB/s	zstd -19 3.3682× decode 167.9 MB/s · comp 3.6 MB/s
Tier 2: M ≤ 5 decode ≥ 33 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s	xz -9e 4.5524× decode 33.6 MB/s · comp 2.3 MB/s
Tier 3: M ≤ 10 decode ≥ 17 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s	xz -9e 4.5524× decode 33.6 MB/s · comp 2.3 MB/s
Tier 4: M ≤ 20 decode ≥ 8.4 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s	xz -9e 4.5524× decode 33.6 MB/s · comp 2.3 MB/s
Tier 5: M ≤ 50 decode ≥ 3.3 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s	xz -9e 4.5524× decode 33.6 MB/s · comp 2.3 MB/s
Tier 6: M ≤ 100 decode ≥ 1.7 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s	xz -9e 4.5524× decode 33.6 MB/s · comp 2.3 MB/s
Tier 7: M ≤ 200 decode ≥ 0.84 MB/s	kanzi -l 9 5.2799× decode 1.0 MB/s · comp 0.8 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s
Tier 8: M ≤ 500 decode ≥ 0.33 MB/s	kanzi -l 9 5.2799× decode 1.0 MB/s · comp 0.8 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.08 MB/s	kanzi -l 9 5.2799× decode 1.0 MB/s · comp 0.8 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s
Tier 10: any speed (no limit)	kanzi -l 9 5.2799× decode 1.0 MB/s · comp 0.8 MB/s	7-Zip -mx9 (LZMA2) 4.6439× decode 40.3 MB/s · comp 10.1 MB/s

28. Archives: ZIP, TAR, TAR.GZ

LZ4 decode speed on this type (M = 1): 372 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 186 MB/s	Brotli 11 1.2833× decode 216.1 MB/s · comp 0.6 MB/s	**HX Fast** **1.2808×** **decode 372.1 MB/s · comp 1.0 MB/s**
Tier 2: M ≤ 5 decode ≥ 74 MB/s	Brotli 11 1.2833× decode 216.1 MB/s · comp 0.6 MB/s	xz -9e 1.2815× decode 100.8 MB/s · comp 2.4 MB/s
Tier 3: M ≤ 10 decode ≥ 37 MB/s	Brotli 11 1.2833× decode 216.1 MB/s · comp 0.6 MB/s	xz -9e 1.2815× decode 100.8 MB/s · comp 2.4 MB/s
Tier 4: M ≤ 20 decode ≥ 19 MB/s	Brotli 11 1.2833× decode 216.1 MB/s · comp 0.6 MB/s	xz -9e 1.2815× decode 100.8 MB/s · comp 2.4 MB/s
Tier 5: M ≤ 50 decode ≥ 7.4 MB/s	**HX Complete** **1.2845×** **decode 10.4 MB/s · comp 0.7 MB/s**	BSC 1.2839× decode 9.1 MB/s · comp 6.2 MB/s
Tier 6: M ≤ 100 decode ≥ 3.7 MB/s	**HX Complete** **1.2845×** **decode 10.4 MB/s · comp 0.7 MB/s**	BSC 1.2839× decode 9.1 MB/s · comp 6.2 MB/s
Tier 7: M ≤ 200 decode ≥ 1.9 MB/s	**H-Deflate + HX Complete + image routes** **1.6446×** **decode 2.4 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.5186× decode 2.6 MB/s · comp 0.6 MB/s
Tier 8: M ≤ 500 decode ≥ 0.74 MB/s	**H-Deflate + HX Complete + image routes** **1.6446×** **decode 2.4 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.5186× decode 2.6 MB/s · comp 0.6 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.19 MB/s	**H-Deflate + HX Complete + image routes** **1.6446×** **decode 2.4 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.5186× decode 2.6 MB/s · comp 0.6 MB/s
Tier 10: any speed (no limit)	**H-Deflate + HX Complete + image routes** **1.6446×** **decode 2.4 MB/s · comp 0.1 MB/s**	precomp (lzma) 1.5186× decode 2.6 MB/s · comp 0.6 MB/s

29. Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.coli), FITS, NumPy

LZ4 decode speed on this type (M = 1): 455 MB/s

Speed range	Best	Second best
Tier 1: M ≤ 2 decode ≥ 228 MB/s	**HX Fast + audio/colour lines** **2.5598×** **decode 250.1 MB/s · comp 0.5 MB/s**	zstd -19 2.4890× decode 353.0 MB/s · comp 3.3 MB/s
Tier 2: M ≤ 5 decode ≥ 91 MB/s	Brotli 11 2.6109× decode 135.6 MB/s · comp 0.4 MB/s	**HX Fast + audio/colour lines** **2.5598×** **decode 250.1 MB/s · comp 0.5 MB/s**
Tier 3: M ≤ 10 decode ≥ 46 MB/s	Brotli 11 2.6109× decode 135.6 MB/s · comp 0.4 MB/s	**HX Fast + audio/colour lines** **2.5598×** **decode 250.1 MB/s · comp 0.5 MB/s**
Tier 4: M ≤ 20 decode ≥ 23 MB/s	7-Zip -mx9 (LZMA2) 2.6388× decode 37.9 MB/s · comp 6.7 MB/s	xz -9e 2.6199× decode 40.0 MB/s · comp 1.8 MB/s
Tier 5: M ≤ 50 decode ≥ 9.1 MB/s	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	BSC 2.6482× decode 10.0 MB/s · comp 10.5 MB/s
Tier 6: M ≤ 100 decode ≥ 4.6 MB/s	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	bzip3 2.6789× decode 8.4 MB/s · comp 8.7 MB/s
Tier 7: M ≤ 200 decode ≥ 2.3 MB/s	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	bzip3 2.6789× decode 8.4 MB/s · comp 8.7 MB/s
Tier 8: M ≤ 500 decode ≥ 0.91 MB/s	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	bzip3 2.6789× decode 8.4 MB/s · comp 8.7 MB/s
Tier 9: M ≤ 2000 decode ≥ 0.23 MB/s	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	bzip3 2.6789× decode 8.4 MB/s · comp 8.7 MB/s
Tier 10: any speed (no limit)	**HX Complete + audio/colour lines** **2.7225×** **decode 11.1 MB/s · comp 0.4 MB/s**	bzip3 2.6789× decode 8.4 MB/s · comp 8.7 MB/s